MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS COBALT BASE, TYPE I, CLASS 2, COMPOSITION C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition(1)

Cr	Ni	Fe	W	C	Mn	Si	Co	P	S
24.5 26.5	9.5 11.5	2.0 max.	7.0 8.0	0.42 0.55	1.0 max.	1.0 max.	Bal	0.04 max.	0.04 max.

(1) Powder shall be reduced by heating in a vacuum or dry hydrogen atmosphere for 2 to 4 hours at 1750° to 1850°F (955° to 1010°C). This requirement shall be waived if powder is produced by atomizing in an inert atmosphere.

Particle Size/Distribution

-170 Mesh (-90 µm)	100.0 min.
-200 Mesh (-75 µm)	95.0 min.
+325 Mesh (+45 µm)	90.0 min.

Sprayed Coating Properties

Bond Strength	41.0 MPa (6000 psi) min.
Hardness	Rockwell 15N, 75 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 15% max.

AMSC N/A FSC 3439
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:31Z
Check the source to verify that this is the current version before use.

Deposit Efficiency and Deposition Rate

65% min for N_2/H_2 @ 15.0 lb/hr
30% min for A/H_2 @ 7.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

Pratt & Whitney Aircraft	PWA 1318
Rolls-Royce, Ltd.	MSRR 9507/3

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/06)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:31Z
Check the source to verify that this is the current version before use.